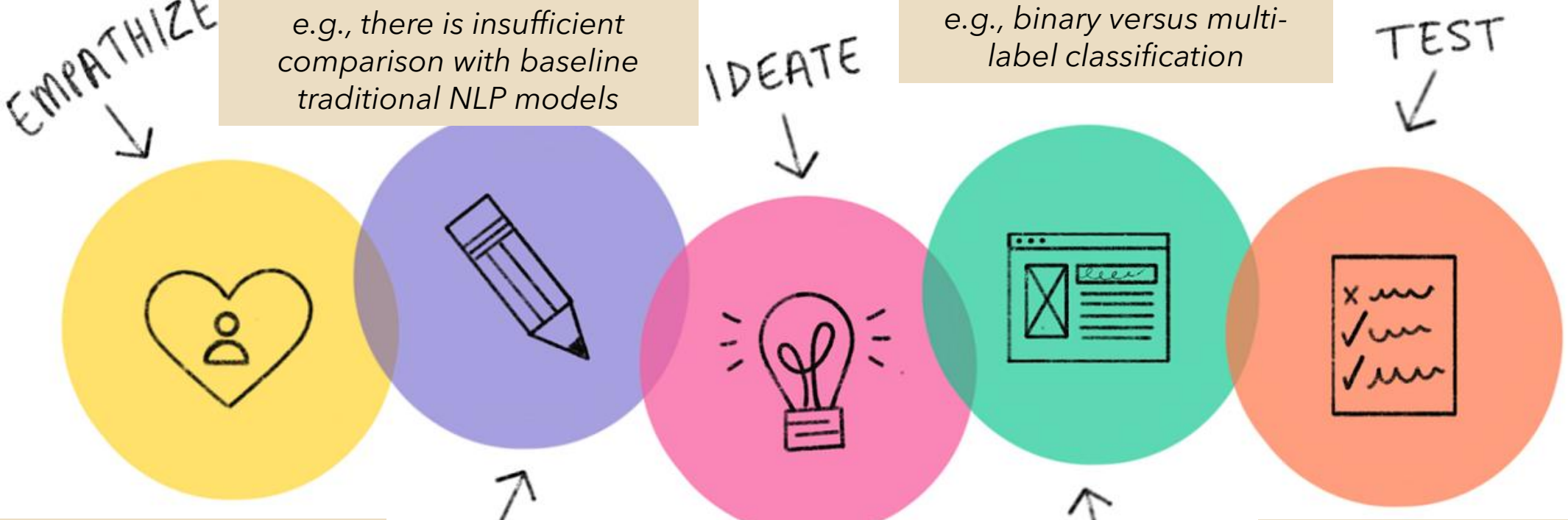
Design thinking overview



What is your problem statement?
e.g., there is insufficient comparison with baseline traditional NLP models

Which datasets are more applicable or suitable to try out the models?
e.g., binary versus multi-label classification

Who is your target audience? What is the purpose?
e.g., want to assist teachers to come up with an automated question generation tool

How do you develop a solution?
e.g., try to implement/ reproduce a recent model or generate insights from variants

How can you compare the performance with several techniques?
e.g., deep learning sequential models versus graph-based networks versus transformers

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Some possible examples for your group project

- **Comparing traditional NLP and deep learning methods for text classification**
  - Week 3 and 4 content: TF-IDF, LDA, word2vec, GloVe (or FastText/ELMo) for feature extraction and then use SVM/ELM/GP to classify text VERSUS
  - Week 5 and 6 content: LSTM, Bi-LSTM, GRU, BERT to perform classification + Week 4 new content: CRF
  - Can apply this for various datasets such as movie genre, question type, news category
  - Mix of binary and multi-class, use precision, recall, F1 score and explain which is better

- **Obtaining better textual features**
  - Week 2 content: Constructing phrases (NP/VP) based on dependency parsing and POS tagging
  - Implement Phrase2Vec/PhraseBERT to show importance of phrases over word/unigram
  - Week 1 content: Analyse differences between n-grams and phrases – what is the impact
  - Can apply this for both short text (such as tweets) versus long text (documents)

- **Compare graph-based approaches against sequential models**
  - Week 5 new content: GCN/GAT/GNN versus Week 5 content
  - Edge weights can be calculated using some of the Week 3 new content on parsing, semantic role labelling, etc.

- **Comparing BERT variants**
  - Week 6 content: BERT, ROBERTA, ALBERT, DISTILLBERT,
  - Explain the differences in architecture with diagrams to illustrate
  - Week 5 content: Impact of hyperparameter tuning, show various plots and how you do optimization
  - Show impact in different applications such as financial news sentiment analysis

- **Seq2Seq downstream tasks (applying transformer models) or developing a chatbot**
  - Week 6 content: Apply different types of LLMs
  - Question answering/Machine translation/Text summarisation
  - Week 4 content: BLEU vs METEOR vs ROUGE, plots

- **Unsupervised learning approaches**
  - Week 4 content: Hierarchical (agglomerative & divisive) clustering versus k-means

- **Exploring new term weighting schemes or new topic modelling techniques**
  - Week 3 in-class activity: e.g., TF-ICF (supervised approach) or Correlated topic models/BERTopic

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Summary of new content/discussion topics so far

Week	Lecture topics	Team activities
2	Word sense disambiguation	Different types of NER, chunking and phrases, generating and interpreting dependency parsing trees
3	Content free grammar, bottom-up (shift-reduce) parsing, coreference resolution, semantic role labeling	Plate diagrams and symmetry, time and space complexity of PCA vs SVD v LDA, various types of term weighting schemes and topic models
4	Conditional random fields	Other word embedding techniques such as FastText and ELMo, different types of machine learning classifiers, hierarchical/fuzzy clustering, impact of evaluation metrics
5	Priority in tuning, graph convolution network, graph attention network	Sensitivity of each hyperparameter, applying deep learning sequential models in different situations
6	Vision transformer	Sensitivity of each component within transformers architecture, design choices, downstream NLP tasks that apply various models and approaches
7	Retrieval augmented generation, chain/tree/graph of thoughts, agentic AI	Leveraging ChatGPT to perform downstream NLP tasks
9	Explainable AI & some LLM tools	-

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Announcement regarding final group project (25%)

Work with your assigned team members

- Refer to rubrics in syllabus for requirements (*e.g., enhance final product with front-end using Streamlit*)
- Consultation session during **Week 12**: bring draft/ideas
- Peer evaluation – will be released later on NTULearn (everybody **MUST** do, else 5% grade deduction)
- Deadline for completion: **Sunday 15 November 11:59pm**

You can choose any (can expand on in-class activities or do something new)

- NLP application (e.g., text summarization, chatbot, sentiment analysis, translation, etc.)
- Datasets (would be good to have 3 or more for comparison)
- Algorithms and hyperparameters for feature extraction and/or computations
- Performance metrics for evaluation
- Baseline papers (conference or journal) for implementation or making slight modifications (*optional*)

Submission mode (one per team, unlimited attempts, last for grading, late: penalty)

- Video max 10min (showing **Python** codes, slides for theory, references) – **no GAI voice, no MATLAB/other**
- Codes (either .ipynb Jupyter notebook or .py files)
- Group reflection document with contributions of each team member and signed
- GAI declaration form (use template)

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Peer evaluation criteria (refer to syllabus)

Criteria (Weighs)	**Score from 1 to 9**[2,3] (**1**: Never; **3**: Rarely; **5**: Occasionally; **7**: Frequently; **9**: Always)				
(For 6-members team)	*Member A*	*Member B*	*Member C*	*Member D*	*Member E*
Member name					
a. Fulfilling one's responsibilities duly (15%)					
Behaved responsibly--such as attend meetings punctually and regularly; participate in discussion; complete assigned tasks/roles punctually.	*Score from 1 to 9*				
b. Fulfilling one's responsibilities effectively (25%)					
Behaved and contributed effectively--such as quality of work produced; creativity of ideas; extensiveness of research and thinking.	*Score from 1 to 9*				
c. Managing interpersonal relationships (30%)					
Listened attentively to and sought inputs from others; helped team resolve conflicts and achieved common understanding to function effectively; promoted respect for others and differences; fostered camaraderie.	*Score from 1 to 9*				
d. Providing support to others to achieve goals (30%)					
Behaved fairly and ethically—such as sharing responsibilities and giving credits. Exhibited group citizenship behavior--such as helping others to learn and complete their work through guidance and encouragement; standing up for others when needed.	*Score from 1 to 9*				

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Purpose and impact

To make sure everyone has a fair contribution

3. Each member's average rating given by his/her team members will be used to determine the final team assignment marks awarded to each member as follows:

Average Rating Range (min: 1; max: 9)	Marks Deduction (out of 100)	Final Adjusted Marks (out of 100 after scaling)
≥ 6.5	0	Original team assignment mark (**M**)
≥ 5.5 to < 6.5	-5	**M** – 5
≥ 4.0 to < 5.5	-10	**M** – 10
≥ 3.0 to < 4.0	-15	**M** – 15
> 1.0 to < 3.0	-20	**M** – 20
= 1.0	**-M**	0

Do the evaluation (ratings and comments) for **ALL** team members

Give honest feedback based on each category (please refer to the syllabus for more descriptions)

Ratings below 6.5 (scale of 1-9; **9 being the best**) will lower all course marks for the person rated

If you do not do it (non-completion) or do it halfway (incompletion), there will be a **penalty in your grades**

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